MACHINE LEARNING ASSESSMENT

Freight Rate Prediction Report

A time-aware gradient boosting approach for predicting posted truckload rates and generating a fixed-lane December 2025 forecast.

Candidate deliverable	Validation predictions, December chart, reproducible code
Training window	2025-01-01 to 2025-10-31 (48,000 labeled loads)
Prediction window	2025-11-01 to 2025-12-31 (12,000 validation loads)
Primary model	HistGradientBoostingRegressor (log-target)
Primary holdout	October 2025 — MAE $135.38 | MAPE 6.41% | R^2 0.819

1. Executive Summary

This report documents the modeling approach used to predict posted freight rates for unseen November–December 2025 loads. The solution is designed around the temporal structure of the data: training ends in October, while scoring occurs entirely in the future. Accordingly, all internal validation uses time-based holdouts rather than random splits.

The final model is a histogram-based gradient boosting regressor trained on a log-transformed target. Distance is the dominant driver of rate, with equipment type, market signals, geography, and calendar effects providing secondary adjustments. Missing values in weight and market index are retained so the model can learn from missingness directly.

Oct Holdout MAE	Oct Holdout MAPE	Oct Holdout R^2	December Mean Rate
$135.38	**6.41%**	**0.819**	**$840.31**

Official Spotter validation metrics are computed after submission. The figures above are internal estimates used to guide model selection and diagnose risk.

2. Problem Statement

Given historical load attributes and posted rates, the task is to:

- Predict **predicted_rate** for every load in the validation set (12,000 unique **load_id** values).
- Produce a 31-day December 2025 rate path for a fixed Lexington → Fort Wayne Dry Van load (360 miles, 32,000 lb), where only the date changes.
- Deliver reproducible code, a validation file, and the official December chart.

3. Data Overview

The labeled training file covers January through October 2025. The unlabeled validation file covers November through December 2025. A separate December template isolates calendar effects on a single fixed lane.

Dataset	Rows	Date range	Role
train-test.csv	48,000	2025-01-01 → 2025-10-31	Labeled training data
validation.csv	12,000	2025-11-01 → 2025-12-31	Unlabeled scoring set
december-chart-inputs.csv	31	2025-12-01 → 2025-12-31	Fixed-lane forecast

3.1 Key observations

- Distance is the strongest correlate of posted rate (approximately 0.91).

- Equipment shifts rate-per-mile: Reefer > Flatbed > Dry Van.
- Weight and market_index contain missing values in both train and validation.
- Validation includes cities not seen in training; coordinates remain available, so geography is modeled with latitude/longitude as well as city categories.
- A small number of extreme loads inflate RMSE; MAE and MAPE better describe typical prediction quality.

4. Validation and Data-Split Approach

Because the scoring window sits entirely after the training window, random cross-validation would overstate accuracy by leaking future market conditions. The evaluation design therefore mirrors the real deployment setting.

Stage	Train window	Evaluation window	Purpose
Primary holdout	Jan–Sep 2025	October 2025	Closest pre-scoring month
Sensitivity check	Jan–Aug 2025	Sep–Oct 2025	Stability across late year
Final model	Jan–Oct 2025 (all)	Nov–Dec 2025	Submission predictions

October is used as the main internal benchmark because it is the nearest complete month to the true scoring period and still contains known targets.

5. Modeling Methodology

5.1 Algorithm

The selected model is **HistGradientBoostingRegressor** from scikit-learn. It handles mixed numeric and categorical inputs, supports missing values natively, and performs well on tabular freight data without heavy preprocessing. The target is modeled as **log1p(posted_rate)** and inverted with **expm1** at inference. Predictions are clipped to a minimum of $50 to satisfy positivity constraints.

5.2 Feature set

- **Load economics:** distance, log(distance), weight, market_index, quote_signal.
- **Geography:** pickup/delivery coordinates; pickup and delivery city categories.
- **Equipment:** Dry Van, Reefer, Flatbed as native categorical features.
- **Calendar:** month, day, day-of-week, day-of-year, week number, weekend flag.
- **Interactions:** distance × market_index and distance × quote_signal.

5.3 Why this design

Distance establishes the base rate level. Market and quote signals adjust that level for tightness and bidding pressure. Equipment and weight capture equipment-specific premiums. Calendar features allow weekly seasonality to appear in the December path without forcing a hand-crafted seasonal formula.

6. Internal Validation Results

The table below summarizes model quality on held-out periods that were never used for the corresponding model fit. The final submission model is refit on all labeled rows after these checks.

Evaluation	MAE ($)	RMSE ($)	MAPE	R^2
Train Jan–Sep → Test Oct	135.38	650.47	6.41%	0.819
Train Jan–Aug → Test Sep–Oct	136.33	640.95	5.92%	0.824
Full-train in-sample (reference)	101.09	552.04	4.36%	0.862

Interpretation: an October MAE near $135 and MAPE near 6.4% indicate that typical loads are predicted within a commercially useful band. RMSE is materially higher because a minority of extreme rate-per-mile outliers remain in the data. Those outliers were retained, since similar cases may appear in the hidden test.

7. December 2025 Fixed-Lane Forecast

The December deliverable holds pickup, delivery, distance, equipment, and weight constant. Only the calendar date changes from 2025-12-01 through 2025-12-31. Because the December template omits market_index, quote_signal, and coordinates, those fields are completed as follows:

- Lexington and Fort Wayne coordinates are taken from the training data.
- Daily market_index and quote_signal are estimated as the mean of validation loads observed on the same calendar date.
- The trained model then predicts one rate per day.

Predicted December rates fall approximately between **$821 and $851**, with a mean of **$840.31**. The series shows a weekly-looking wave, consistent with calendar and market variation rather than a flat constant. Historical Lexington → Fort Wayne Dry Van loads in training occupied a similar band (roughly $760–$970).

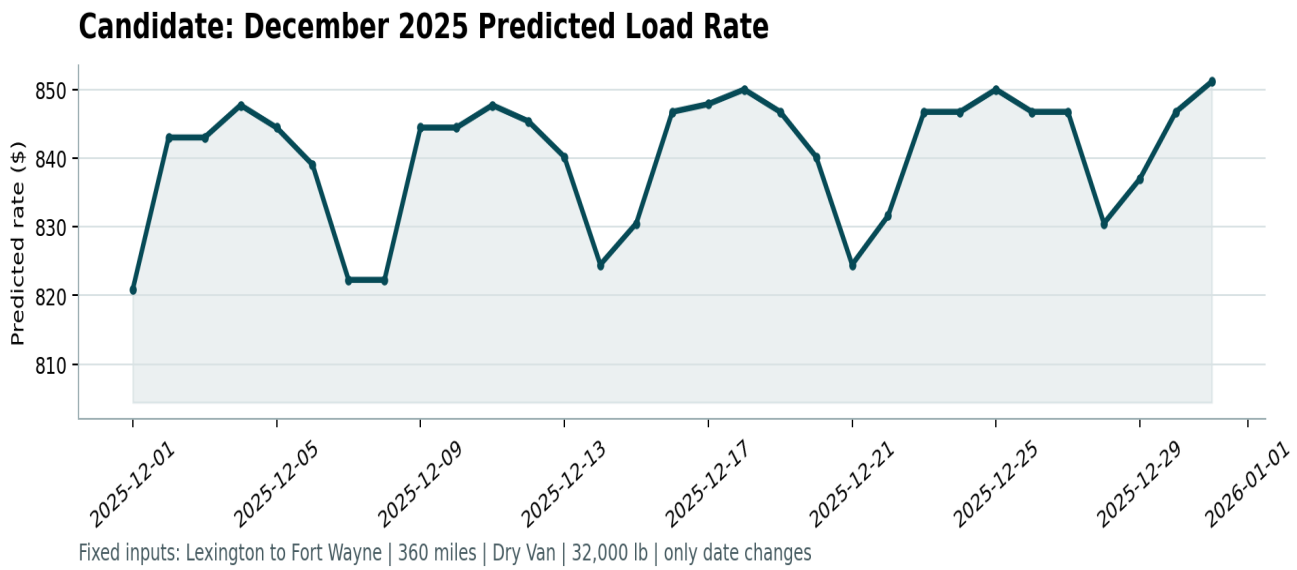


Figure 1. Official scorer chart — Candidate December 2025 predicted load rate (Lexington → Fort Wayne | 360 miles | Dry Van | 32,000 lb).

8. Reproducibility

All outputs can be regenerated from the repository with the following commands:

Step	Command
1. Install dependencies	`python -m pip install -r requirements.txt`
2. Train and predict	`python train_predict.py`
3. Validate + chart	`python score.py --predictions validation_predictions.csv --december-predictions data`

Primary outputs: **validation_predictions.csv** (12,000 rows), **december_chart_inputs.csv** / **data/december_chart_inputs.csv** (31 rows), and **scorer_results/candidate_december.png**.

9. Closing Notes

This submission prioritizes a deployment-realistic validation design, transparent feature choices, and fully reproducible artifacts. The model captures the dominant distance effect while allowing market, equipment, geography, and calendar signals to refine predictions for both the general validation set and the fixed December lane.

Future improvements, if additional iteration were available, could include LightGBM/XGBoost comparisons, time-safe lane encodings, and holiday-aware calendar flags. The current pipeline already satisfies the assessment deliverables and passes the official local scorer checks.